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Survey Paper

Recent advances in vision-based indoor navigation: A systematic literature review☆

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ABSTRACT

Indoor navigation has remained an active research area for the last decade. Unlike outdoor environments, indoor environments have additional challenges, such as weak signals, low light, and complex scenarios. Different technologies are used for indoor navigation, including WiFi, Bluetooth, inertial sensors, and computer cameras. Vision-based methods have great potentials for indoor navigation as they fulfill most of the general requirements such as minimal cost, ease of use, ease of implementation, and realism. Therefore, researchers have successively proposed different novel vision-based approaches for indoor navigation. Unfortunately, there is no standard review article (except a few general reviews) that covers the current trends and draws a pipeline for future research. In this paper, we reviewed the current state-of-the-art vision-based indoor navigation methods. We followed the systematic literature review (SLR) methodology for article searching, selection, and quality assessments. In total, we selected 68 articles after final selection using SLR. We classified these articles into different categories. Each article is briefly studied for information extraction, including key idea, category of the article, evaluation criterion, and its strengths and weaknesses. We also highlighted several interesting future directions. This study will help new researchers to grasp the research challenge as well as present the results of their research in the field. It will also help the community to find a suitable indoor navigation system according to users' requirements.

1. Introduction

Indoor navigation has been gaining a key attention of researchers, especially in the past decade. A rapid growth of amazing technologies has been found in this field. General applications of such technologies include human assistance (visually impaired or aged people), robot navigation, tourists' guidance, augmented reality games and training [1–3]. Visually impaired people (VIP) needs special assistance such as electronic devices that are preferably low-cost, simple to interact with, easy to carry on, and provide accurate information [4]. The state-of-the-art technologies are found in three major categories including computer vision-based methods, wireless technologies, and dead reckoning technologies. Computer vision-based methods are also found in two different categories including markerless or marker-based methods [5]. Fiducial markers are printed images which are recognized with video camera. Various toolkits [6–10] are used for generation and recognition of Augmented Reality (AR) markers. The use of sensors such as accelerometer, compass, gyroscope, and magnetometer is the key filler in the dead reckoning techniques. Different alternative technologies for indoor navigation include wireless techniques [11,12], near field communication (NFC) [13], infrared (IR), radio frequency identification (RFID) [14–16], bluetooth low energy (BLE) beacons [3], graph-based techniques [17], graphical models [18], point clouds [19] and bluetooth/WiFi [20–22].

Considering its accuracy, ease of use and installation, and low-cost, computer vision-based methods are getting key attention of researchers [23]. Global Positioning System *(GPS)* is considered a de facto and ideal solution for outdoor positioning and localization [24]. However, it often fails in indoor navigation. Generally, users prefer an indoor navigation system that is low-cost, easy-to-use, easy-to-maintain, and is consuming low-power [25]. Unlike other alternatives [11–16], vision-based methods are comparatively more user friendly as these methods fulfill the above-mentioned general preferences. Moreover, since the camera acts

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as an artificial eye, the vision-based methods are more realistic especially for visually impaired people (VIP). Therefore, computer vision is an optimal solution for indoor navigation. There are different computer vision-based methods used for indoor navigation. However, there is no significant work in the context of survey papers. There are only a few review articles on general indoor navigation. However, there is no such article available to provide a comprehensive review of the state-of-the-art computer vision-based methods for indoor navigation. Furthermore, none of the existing review articles use any standard methodology for article searching, selection and data extraction. Therefore, a comprehensive review on the computer vision-based methods is required to guide the new researchers in the field and to help general users in finding suitable methods easily.

Systematic literature review (SLR) [26] is a methodology used for reviewing state-of-the-art articles on a given topic in an unbiased manner by following a predefined set of rules for articles searching, articles selection, and data extraction. SLR is mostly used in software engineering and medical fields. To the best of our knowledge, even though there is a rapid growth of research on vision-based indoor navigation, there is no SLR on the topics (see Section 1.3) to provide a comprehensive sketch of the state-of-the-art methods. In this paper, we imitate SLR methodology [26] from the software engineering field to review the state-of-the-art articles on computer vision-based methods for indoor navigation. We used a pre-defined strategy for searching articles, a criterion for selecting articles, an analysis mechanism for assessing article quality, and a strategy for extracting data from each article. In total, we selected 68 articles for data extraction. We divided the articles into six different categories. Each article is briefly described with its key idea, pros and cons, evaluation methods, input source (sensors), output data (feedback), and final achievements. In addition, we highlighted several interesting directions in indoor navigation for future research.

1.1. Research background and motivation

Vision-based indoor navigation has also been getting the key attention from researchers. Fig. 1 illustrates different positioning systems and ways of providing visual guidance. Indoor navigation systems can help not only visually impaired people but also normal users, especially in unknown environments. However, there is not yet a review article to collect all these articles and provide a comprehensive analysis of their data and experimental results. Although there are few review articles in indoor navigation (as described in Section 1.3), still these review articles are either very general or focused on other sub-domain instead of vision-based methods (i.e. our current focus). Inspired by the SLR methodology [26] from the software engineering field, and its adaptation in other fields [27,28], we aim to collect all existing articles on vision-based indoor navigation using SLR methodology. The study will not only provide a comprehensive review of existing methods for new researchers, but also provide different ways and metrics for results analysis. Furthermore, a number of research challenges highlighted in this paper can be considered as future research.

1.2. Vision-based indoor navigation and its challenges

Unlike existing survey articles (Section 1.3), our review is novel in two aspects. First, we followed SLR as a standard methodology for review. Second, our focus is also narrow on vision-based methods for indoor navigation. In our context, the term **indoor navigation system** can be defined as a system that utilizes computer vision techniques (or camera) for tracking or providing visual feedback to users. It helps human beings (or even a machine/robot) in navigation from one place to another place in a given indoor environment. User localization, i.e., finding a user location in an indoor environment, path planning, and guiding users are factors in a typical indoor navigation system. Obstacle avoidance is also a key consideration, especially for blind users or robots. Eq. (1) shows that the navigation system is composed of tracking and user localization, and planning while Eq. (2) shows that planning is composed of optimization (finding optimal path), rendering, and obstacle avoidance.

$$Navigation = Tracking + Localization + Planning, \tag{1}$$

where,

$$Planning = Optimization + Rendering + Obstacles. \tag{2}$$

Unlike outdoor environments, indoor navigation is more challenging. Considering the literature [1,32], we have listed a few key challenges of vision-based indoor navigation.

(1) **Accurate, robust and continuous functioning:** It includes increasing accuracy, robustness and reliability and decreasing errors. Since vision-based methods are typically affected by light intensity and occlusion; the accuracy is a challenging task in this context.
(2) **Adding new features and scaling:** Most of the existing methods are either specific to a particular building or difficult to extend with new features. Therefore, proposing a generic, extendable, and easy to install system is a key challenge.
(3) **Signal strength:** Generally, indoor environments are found with weak signals. Vision-based methods may have some hybrid configurations where the signal strength is an issue.
(4) **Computational cost and efficiency:** Vision-based methods work with real-time pattern recognition, which needs optimal memory storage policy and efficient processing.
(5) **Localization of moving users:** Since the user is in continuous motion, it is challenging to recognize a point of interest or marker for user localization, and thereby false-negative rate can be increased.
(6) **User comport and difficult to use:** Although, vision-based systems provide more information than other alternative systems. Still, the users feel fatigued and socially in partial cut off from the real world. Another challenge is the rapid change in the indoor scene (such as wall painting and decoration) may lead to change the pre-defined point-of-interests.

Similarly, some issues generally faced by any indoor navigation system are also found with vision-based systems. These include the installation cost (money, time, space, weight, and energy), complexity (software, and hardware), and robustness [1,33].

The key advantages of the vision-based method over other alternative methods are listed below.

1. Computer camera is real imagination of the human eye, and visual information is easy to understand. Therefore, vision-based methods are easy to navigate as compared to other methods.
2. Since most devices and smartphones have cameras so these are low-cost. The power consumption is also lower.
3. There is no (or lower) chance of information delay.

The key limitation of the vision-based system is immunity to light intensity and its direction. In addition, the proper placement of the marker (or other target tracking object) is necessary. Moreover, the first-time installation might be time-consuming in some cases.

We have tried to narrow our study direction to vision-based methods only. The detailed criteria for articles selection (according to the scope of our study, and quality of the existing article) is given in Section 2.3.

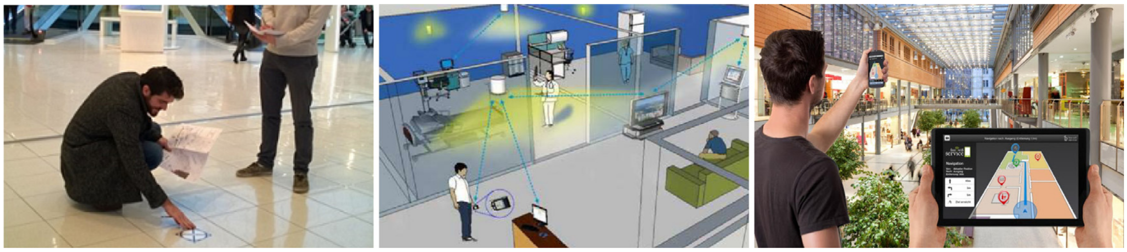

Fig. 1. Indoor navigation systems: from left to right: visual landmarks for indoor navigation [29], Wifi, RFID and Blutooth based system [30], and visual directions for navigation [31].

1.3. Existing survey articles

With the key attention of researchers, several research articles have been published time by time. In the same way, a number of review articles on indoor navigation, localization, and position tracking have also been published. In this section, we will briefly describe a few relevant studies in the context of our current study. The related studies include a survey on mobile systems for indoor navigation [34] which provides an evaluation framework for the analysis of existing mobile systems for indoor navigation. Further studies include a review of fingerprinting-based and mobility tracking techniques [35], a survey of calibration-free methods [36], pedestrian indoor positioning systems [37], navigation via visible LED lights [38], and general reviews on indoor navigation, existing challenges, and animal tracking in the indoor environment [39,40]. Similarly, the theoretical approaches and applications have also been explored [41]. However, none of these survey papers have used the SLR methodology.

Gu et al. [42] conducted a review of indoor navigation in wireless personal networks with main focus on commercially available and research-oriented systems. The term personal network is defined as a network of different personal devices in the indoor environment such as chairs, tables, beds, etc. The review article [42] has given a comprehensive view of the stat-of-the-art and highlighted different key factors to be considered while designing indoor navigation systems. However, the study is 11 years old and it is not as specific as our current study. Another survey article [43] is very close to our current study. However, this article [43] is concerned with mobile robot navigation and was published 19 years ago. Simultaneous localization and mapping (SLAM) technology [44] has also been reviewed, however, it is mainly focused to robot navigation and also specific to SLAM. Another recent survey article [45] is very close in direction with our current study. They have briefly explored computer vision-based methods for indoor localization with advantages and limitations. However, they have not adopted any standard method. Furthermore, their focus is mainly on localization rather than navigation.

Recently, a meta review [46] has been carried out which provides an analysis of 62 recent survey articles on indoor positioning systems. It also provides an analysis of the articles being cited in the selected surveys. Another recent survey [47] has reviewed a broad domain of indoor position and pathfinding methods with detailed discussions. It reviews the papers in different categories such as computer vision-based methods, communication technologies used for indoor navigation, and evaluation methods adopted in the literature. Other surveys for indoor navigation/positioning include machine learning methods [48,49], wireless methods [33], cellular localization systems [50], inertial sensors [51], Optical indoor positioning [52], context-aware navigation methods [53], navigation methods for the visually impaired [40,54,55], Channel state information (CSI) based methods [56], Wifi and fingerprint-based method [57], and generic reviews [58], and smartphone sensor based navigation [59]. However, none of these have used some standard methodology like SLR. Furthermore, there is no good survey on the vision-based methods. Perhaps there are four systematic literature reviews on indoor navigation. One of them focused on heuristics-based methods [28], which has selected 93 primary studies. However, being limited to heuristics information, this survey [28] has no specific information on vision-based methods. The second one [60] focused on mobile applications and compared only 15 applications. Among these 15 articles, they found 5 for indoor navigation, 3 for outdoor, and 8 on both. A third SLR [61] has also been done on a related topic. However, it is for outdoor navigation with the focus on blind users. Moreover, it [61] is not specific to computer vision. The fourth SLR [62] is recently published on collaboration indoor positioning systems. However, there is no SLR on vision-based indoor navigation.

A recent review article [63] has briefly described the existing methods with two major concerns including indoor positioning and shortest path finding. However, the study [63] included only a few articles while missing many interesting articles in the field. Zafari et al. [64] conducted a survey with considerations on generic indoor localization and focused specifically on the internet of things technologies for indoor navigation. Though this survey [64] provides a significant exploration on the topic and presents pros and cons of the stat-of-the-art; still it has two limitations. First, they have not used any standard method like SLR to avoid possible bias. For example, they have more focused on IoT technologies in their study. Second, their study focus was to review all the indoor localization techniques rather than a narrow direction. In summary, our current review is novel in its methodology, narrow being focused on the vision-based method and significant being the first SLR on the topic. Unlike all these existing methods, we follow a standard methodology (i.e. SLR) to review all existing research in a narrow direction i.e. vision-based navigation. Moreover, we provide quantitative measures for different parameters such as the quality of the paper, and evaluate the existing research article to analyze their way of results evaluation, the number of users in the user study and other parameters. These parameters can help junior researchers to find an existing issue and to learn how the results are analyzed in this field.

1.4. Contributions

In summary, we have the following contributions.

1. We conducted the first SLR in vision-based indoor navigation. We followed a predefined protocol in each phase of the review including article searching, article selection, data extraction and results analyses. We selected 68 primary studies after final selection via SLR.
2. To classify the SLR findings and to provide a concise understanding of the facts, this study classifies the selected articles into six different categories and briefly described the strengths and weaknesses of each article.

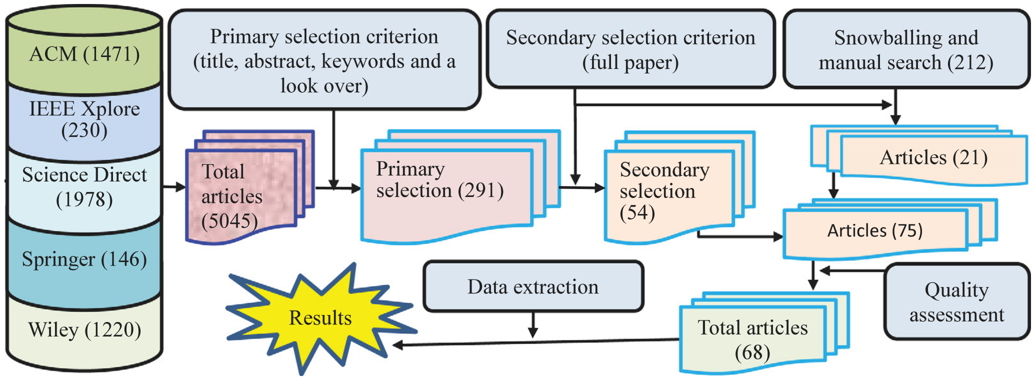

Fig. 2. Research methodology, including search strategy, article selection criteria, quality assessment, and data extraction.

3. We provided a short description for each method. The tools/sensors used, nature of the feedback, evaluation mechanism, the number of users in the user study, and different other important parameters are extracted from each article.
4. We analyzed the quality of each article and calculated a quantitative score indicating quality of each article. The calculated score is depending on four different quality metrics (see Section 2.4).
5. We identified several important research issues on the topic and identified the state-of-the-art methods used for evaluations in indoor navigation. These findings will contribute to further research in the community.

2. Review methodology

Inspired by the popularity of SLR, we followed the SLR methodology [26] and defined a set of rules for all phases of the review. An SLR is a systematic procedure for identifying, analyzing, and interpreting the related published studies with reference to particular research questions. It is an approach, where research questions are designed, which can be critically analyzed through data gathered from the studies identified during the review process as a sample [65]. It evaluates the threat of bias and quality of evidence in individual studies [66]. In SLR, the criteria for article searching, results gathering and analyses, article quality assessment, and article selection are decided at start of the review and are strictly followed throughout the review process [26,27]. Fig. 2 illustrates the overall pipeline of our study which is further described in the following subsections. SLR is mostly used in the software engineering field, however, it is extending time by time to other fields [27] as well.

2.1. Formulating research questions

Considering our study objectives, the following research questions were used in the study.

RQ 1: What are the challenges of computer vision-based techniques for indoor navigation?
RQ 2: What are the computer vision-based techniques used for indoor navigation?
RQ 3: What are the advantages and limitations of the state-of-the-art computer vision-based techniques used for indoor navigation?
RQ 4: What are the metrics used for the analysis of the computer vision-based indoor navigation techniques?

2.2. Strategy for searching articles

The way we search for relevant articles from their online libraries is also important. We designed a search string according to the SLR guidelines [26] and selected five different libraries for searching the related articles. The keywords and their synonyms (form our research questions (Section 2.1)) were concatenated using Boolean operators for making the search string. The finalized search string used in our study is given below.

(vision AND indoor AND navigation AND (People OR Human OR visitor) AND (path OR Location OR Localization))

We used five different libraries (as shown in Fig. 2) for articles search using above string. In addition, we used manual search, and the snowballing strategy [67] for searching further articles. Snowballing includes forward snowballing i.e. searching for relevant articles cited in a given article and backward snowballing i.e. searching for relevant articles citing a given article.

2.3. Criteria for article selection

The selection criteria were followed in two phases. In the primary selection phase, we read the title, keywords, and abstract of each article and included relevant articles. In the secondary selection phase, we read all articles and excluded articles based on the following inclusion and exclusion criteria:

1. The articles specific on vision-based indoor navigation were included, and the articles not using vision-based methods were excluded.
2. Peer-reviewed journal articles and conference papers were included and non-peer-reviewed reports and books were excluded.
3. Articles presented in English were included and those written in other languages were excluded.
4. Articles duplicating other articles were excluded.
5. We conducted a quality assessment (Section 2.4) and excluded articles with assessment scores less than 2.

2.4. Quality assessments of articles

In addition to the above inclusion and exclusion criteria, we analyzed each article based on its quality, contents, and venue of publication. Similar criteria were used in a recent SLR article [27]. The article quality was calculated numerically, from 1 to 4, by answering the following four questions with numeric values. The four questions are listed below.

1. How many are the annual citations of the article (number of citations per year using Google Scholar citations)?
2. Is the article published in a good venue (a relevant and high-quality journal/conference)?
3. Is the method novel, appropriate, and helpful to the community? Is it easy to implement?
4. Are the results analyzed appropriately (i.e. an unbiased user study and/or some other analysis method)?

The answer for each question was selected one of the three options, i.e. “good: 1”, “average: 0.5”, or “poor: 0”. The numerical values of all answers were added to calculate the accumulative assessment score for each article. Articles with 2 or more accumulative scores were included and the remaining were excluded. Question 1 was answered as 1 for recently published papers (published after 2018) and also for articles having 5 citations per year. Similarly, articles with yearly citations ≥ 3 were answered as 0.5 and the remaining were answered as 0 for question 1. Question 2 was answered by checking quality and relevance of the publication venue (Journal/Conference). Articles from high-quality and relevant journals/conferences were answered as 1 and the remaining articles from multidisciplinary lower reputation venues were answered as 0.5 or 0. Similarly, questions 3 and 4 were also answered after a careful reading of the articles, and their contributions, implementations and fairness and completeness of the results. We excluded 7 articles in the quality assessments phase.

2.5. Data extraction and synthesis

Based on our research questions, we extracted the following information from each selected article:

1. The key issues and the proposed solution in the selected article.
2. The key algorithm, its advantages and disadvantages, and possible future directions.
3. The way the results are analyzed and presented, and the number of users in the user study (if any).

Next to data extraction, we analyzed the collected data. Considering the nature of the indoor navigation system in each article, we put the article in a separate category. In this way, we classified the articles into six different categories. The collected information is presented in these six categories. The way each article is analyzed and the system in the article is evaluated are presented in a separate section (i.e. Section 5).

2.6. Data presentation and paper organization

In the previous section, we introduced the scope, background, and importance of our current study. An overview of related survey articles and a summary of the key contributions are also given. This section (Section 2) is giving a summary of the research methodology adopted. Considering our research questions, and the extracted data, the remaining of the paper is organized as follows.

1. Sections 3, and 4 provides method based and system based classification of the selected articles, respectively. The results are summarized in Table 1 answering *RQ 1*, and *RQ 2*. Similarly, Table 2 provides pros and cons of each article (i.e. *RQ 3*).
2. The data extracted for *RQ 4* i.e. the metrics used for the analysis of vision-based indoor navigation systems and the available datasets have been presented in Section 5.
3. Section 6 provides few interesting future research directions, while Section 7 provides discussions and concluding remarks.

3. Indoor navigation methods

Vision-based indoor navigation methods are found in two main categories including marker-based and markerless methods. The indoor navigation systems can further be classified into four different categories (as presented in Section 4). Fig. 3 shows our classification of the articles and the overlap among different categories. Though it is difficult to find a good categorization, we presented the data extracted in two different categories of methods in this section. Further results have been presented in Section 4. Table 1 shows a brief summary of the extracted information from each article, whereas the articles' quality assessment scores, and pros and cons are given in Table 2.

3.1. Marker-based methods

Camera tracking is a key challenge in Augmented Reality applications [134]. Camera tracking which is achieved in either markerless or marker-based methods can be used for user localization. In marker-based tracking, printed markers are placed in the indoor environment at different places such as floor, ceiling, and walls. These markers are traced by a video recording camera and are used for location tracking and path planning. There are different toolkits such as Vuforia [135], ARToolKit [7], ARToolKit Plus [9], ARTag [6], AprilTag [136], and ArUco (Augmented Reality University of Cordoba) [10]. Marker-based tracking techniques are more robust and work under different light conditions as compared to markerless techniques [83]. However, marker-based systems are still affected by light conditions and the direction of light source [137,138]. Each of these toolkits has its advantages and disadvantages. For example, a comparative study [139] indicates that Vuforia has good performance over ArUco and AprilTag. However, Vuforia is not open source whereas ArUco and AprilTag are open source. Though the current trend is markerless tracking, there are still very interesting articles on marker-based tracking.

Khan et al. [1] have used ARToolKit markers [7] for indoor navigation. Their method is low-cost as it uses only cameras. Furthermore, their method is generic and can be easily installed in any indoor environment. However, there are two main issues in their current study. First, the placement of physical prints is an issue and sometimes the building administrator might be unhappy with marker placement. Second, the markerless method or the use of hidden markers can be future work. Furthermore, a handheld smartphone is difficult to carry while navigating in the indoor environment in their method. A similar method was proposed in 2011 which used a laptop camera for the detection of ARToolKit markers [68]. It shows the directions using AR overlay and audio sounds. The main issue with this method [68] is carrying a laptop during navigation, and lack of extendibility and generalization (i.e. installation in other environments). Furthermore, artificial intelligence and deep learning can be applied as future work. Zeb et al. [69] proposed a similar approach that can be used by visually impaired people for indoor navigation. However, it also has the same limitations of carrying a laptop and the lack of generalization.

Similarly, Kim et al. [70] have also used ARToolKit markers for location information and indoor navigation. They developed an application that transmits image sequences to a remote server for marker detection, which is used to calculate the user's position. However, there is no mechanism for shortest path finding. For results analysis of this paper [70], the authors have analyzed different modules of the proposed method. For example, analysis of marker detection, image matching module, and location recognition with the model was performed. However, unlike other typical methods, there was no evaluation of user navigation via a user study. A previous paper [71] has used a similar idea with

Table 1
A brief summary of all articles included after final selection.

Paper ID	Method name/short title	Sensors	Feedback	Key issues addressed	No. of users	Category	Year
S01 [1]	Generic smartphone-based Indoor navigation	Camera	Audio, Visual	Low-cost and robust tool	10	Markers, Smartphone	2019
S02 [68]	AR based indoor positioning navigation tool	Camera	Audio, Visual	Low-cost	–	Markers using Laptop	2011
S03 [69]	Indoor vision-based auditory assistance for blinds	Camera	Audio	Low-cost	05	Markers using Laptop	2011
S04 [70]	Vision-based location positioning using AR	Camera	Audio,Visual	Robustness and low-cost	–	Markers using Laptop	2008
S05 [71]	Structured visual markers for indoor pathfinding	Camera	Visual	Low-cost solution	–	Markers using Laptop	2002
S06 [72]	Headloc	Camera	Audio	HMD i.e. Google glass	08	Wearable technology	2014
S07 [73]	Ebsar: Indoor guidance for VIP	Camera, Sensors	Audio	HMD i.e. Google glass	24	Wearable technology	2016
S08 [74]	AR-based indoor navigation using google glass	Camera, Sensors	Visual	3D point cloud and Google glass	27	Wearable technology	2016
S09 [75]	*ViNav:* Vision-based indoor navigation	Camera, Sensors	Visual Map	Meshes; efficient, scalable	13	Hybrid, vision + sensor	2019
S10 [76]	Indoor navigation system using visual positioning	Camera	Visual	Low-cost	–	Markers using smartphone	2018
S11 [77]	INSAR	Wifi, Camera	Visual	AR visualization for navigation	7	Hybrid, AR visualization	2014
S12 [78]	Smart indoor positioning/Location and Navigation	Camera	Visual	Use NFC and QR code	–	Smartphone-based method	2013
S13 [79]	STEPS: An indoor navigation framework	Camera, Sensors	Visual	Simple yet efficient AR Technology	–	AR Technology	2020
S14 [80]	Hybrid indoor localization (IMU and Smartphone)	Camera, Sensors	Visual	Improving accuracy	01	Hybrid (Camera + sensors)	2020
S15 [81]	Indoor positioning via static objects recognition	Camera	Visual	Large environment	–	Deep learning, smartphone	2018
S16 [82]	Visual positioning indoors: Human eyes vs. Cameras	Camera	Visual	Improved accuracy	10	Smartphone based system	2017
S17 [83]	Indoor positioning and navigation with camera	Camera	Visual	Low-cost solution	20	Smartphone and markers	2009
S18 [84]	RGB-D camera based wearable navigation system	RGBD Camera	Tactile	RGB-D camera	10	Wearable technology	2016
S19 [85]	ISANA	RGBD camera	Audio	Context Awareness	–	Wearable technology	2016
S20 [86]	Vision-based mobile indoor assistive navigation	RGBD Camera	Audio, Haptic	Motion estimation, Robustness	04	Wearable technology	2018
S21 [87]	An ARCore based navigation system	Camera, Sensors	Audio, Haptic	Optimal mapping, Fluent guidance	–	AR Technology, smartphone	2019
S22 [88]	WiFi and Vision-integrated self-localization	Camera,Wifi	Visual	Robust tracking	–	Smartphone	2020
S23 [89]	ARBIN	Camera, Sensors	Visual	Accuracy and visual directions	04	Markerless, via smartphones	2020
S24 [90]	Augmented-reality-based indoor navigation	Camera, Sensors	Visual, Audio	Efficient and effective navigation	39	AR, Smartphones, HMD	2017
S25 [91]	Wearable travel aid for navigation	RGBD Camera, IMU	Audio	Avoid collision, Effective navigation	20	Wearable technology	2019
S26 [92]	Blind guide– A virtual eye	Stereo camera	Audio	Easy navigation	–	Wearable technology	2014
S27 [93]	A Fusion of vision and depth sensor	RGB-D camera	Audio	Efficiency, Robustness	02	Wearable technology	2015
S28 [94]	Recovering the sight to blind people	Camera,IMU	Audio	Multiple sensors	–	Marker tracking	2016
S29 [95]	Enabling independent navigation for Vis. Impaired	Camera	Audio,Haptic	Finding free space	15	Wearable technology	2017
S30 [96]	Navigation assistance using RGB-D sensor	RGB-D camera	Audio	Low-cost solution	–	Markerless technique	2016
S31 [97]	Indoor navigation using landmarks and a GIS	Camera	Audio	Integration GIS and landmarks	–	Markerless technique	2012
S32 [98]	Co-robotic cane	RGB-D camera	Auto-steering	Cane with robotic functions	–	Markerless technique	2016
S33 [99]	Vision-based localization for indoor navigation	Camera	Audio	Text recognition	–	Smartphone-based	2014
S34 [100]	Indoor navigation interface for vision-based localization	Camera	Visual	VR,AR	81	Smartphone-based	2012
S35 [101]	Indoor robot navigation via siamese DCNN	Camera	map	Deep learning	–	Robot navigation	2018
S36 [102]	Comparative study on BLE vs Computer vision	Camera,BLE	Audio	Comparative study	10	QR marker	2019
S37 [103]	Toward a computer vision-based wayfinding aid	Camera, Audio	Audio	Text and door recognition, SVM	–	Markerless	2013
S38 [104]	Indoor navigation using CAMShift and D* algorithm	Camera	Audio	Object recognition	03	Markerless	2016
S39 [105]	An efficient indoor navigation using QR codes	Camera	Audio	QR Code	–	Smartphone	2015

(*continued on next page*)

Table 1 (*continued*).

Paper ID	Method name/short title	Sensors	Feedback	Key issues addressed	No. of users	Category	Year
S40 [106]	Indoor navigation using a digital sign system	Camera	Audio	Tag detection	–	Smartphone	2013
S41 [107]	Vivid: augmenting navigation with edge computing	Camera	Visual	Edge computing	36	Smartphone	2020
S42 [108]	Vision-based system for blind people to wander	Stereo camera	Audio, Haptic	3D Segmentation	10	Wearable systems	2021
S43 [109]	CamNav: a computer-vision indoor navigation system	Camera	Visual	MSLBP features, DL	–	Mobile Devices	2021
S44 [110]	Indoor localization for visually impaired travelers	Camera	Audio	Exit sign recognition	05	Smartphone	2020
S45 [111]	Navigation using computer vision & Sign Recognition	Camera	Audio	Exit sign recognition	–	Smartphone, Markerless	2020
S46 [112]	A low-cost tracking system for mobile robots	Camera	Robot signals	ArUco Markers	–	Marker tracking	2020
S47 [113]	Sensor fusion localization and navigation for VIP	Camera, sensors	Vibration	Camera detection	–	Hybrid, Smartphone	2018
S48 [22]	Camera-based multimodal indoor localization	Camera, Wifi	None	Visual detect + Wifi	08	Hybrid, Smartphone	2020
S49 [114]	Visual positioning system for navigation	Camera	Audio, Visual	Marker Tracking	–	Markers	2017
S50 [115]	Indoor navigation for shopping complexes	Camera	Audio, Visual	Signboard Tracking	10	Smartphone	2020
S51 [116]	Indoor navigation using annotated maps in AR	Camera	Visual map	ARCore	15	Smartphone	2019
S52 [117]	Feature-based indoor navigation using AR	Camera	Visual	Features	20	Markerless, Smartphone	2013
S53 [118]	Image sensor based navigation system	Camera	LED lights	LED transmitters	–	Smartphone (mobile)	2018
S54 [119]	Travi-Navi: Self-Deployable Indoor Navigation	Camera	Visual	Images + Sensors	04	Mobile Device	2017
S55 [120]	Mobile robot indoor dual Kalman filter localization	Stereo Camera	Robot Signals	Stereo vision + Sensors	–	Robot	2017
S56 [121]	Vision-aided robot's indoor navigation	Two Cameras	Robot Signals	Two Cameras Implementation	–	Robot	2016
S57 [122]	Virtual-blind-road based wearable navigation	Camera, Sensors	Audio, Visual	Wearable	–	Wearable Technology	2018
S58 [123]	Smart guiding glasses for VIP in indoor environment	Camera, Sensors	Audio, Visual	Wearable	20	Wearable Technology	2017
S59 [124]	An indoor navigation system for the VIP	Infrared Camera	Audio	Usability Enhancement	05	Markerless, Smartphone	2012
S60 [125]	Infrared & Camera Fusion for Indoor Positioning	Infrared, Camera	–	Robustness	–	Hybrid	2019
S61 [126]	Indoor navigation for VIP using AR markers	Camera	Audio	Low-cost	–	Marker, smartphone	2017
S62 [127]	MOMA: visual mobile marker odometry	Camera	Robot signals	Low-cost	–	Marker, robot	2018
S63 [128]	Optimal landmark deployment for indoor localization	Camera	Robot signals	Low-cost, Simple	–	Marker, Robot	2017
S64 [129]	QRPos: indoor positioning with QR codes	Camera	Robot signals	Low-cost	–	Marker, Robot	2016
S65 [130]	AR-based navigation using RGB-D camera & map	Camera	Visual	Depth Camera, Error correction	–	Markerless	2021
S66 [131]	Indoor positioning with a single camera using PnP	Camera	Visual	Perspective-n-Point (PnP)	–	Markerless	2015
S67 [132]	Indoor navigation on wheels (and on foot)	Camera	Visual	Speed detection	11	Smartphone	2012
S68 [133]	Visual odometry for indoor navigation	Camera	–	Visual SLAM	–	Smartphone	2012

a laptop hung on the back of the user aided by a camera and an inertial tracker on the user's head. It used ARToolkit markers for user tracking and the guidance information was displayed on a wrist-mounted touch screen. This method [71] provided considerably accurate results under normal light conditions but is bulkier because of carrying a laptop. Furthermore, it requires manual editing of map coordinates instead of automatic map generation which has been achieved in the recent research [1].

Yadav et al. [76] also used ARToolKit markers for indoor navigation. However, there is only visual output for navigation and there is no audio like other ARToolKit based methods [1,69]. Furthermore, there is no scientific escalation of the method [76] so that the reader can obtain more information about it. Yang et al. [126] used ArUco markers [140] and followed MILP (Mixed Integer Linear Programming) approach for marker tracking.

A comparative study of BLE and computer vision [102] has recently been carried out. They developed and compared three methods including two vision-based methods (i.e. a CNN-based method and quick response code (QR-code) based method) and one BLE based system. The vision-based systems were implemented on a laptop and the BLE based system used smartphones. Their results reveal that the vision-based method is more effective in terms of accuracy and usability. Another method [114] also used QR markers for indoor and outdoor navigation. This method [114] provides textual, audio, and visual maps as feedback to the users.

Table 2
Advantages and limitations of the state-of-the-art methods. *Quality* is a numerical quality assessment score for each article.

Paper ID	Advantages	Limitations	Quality
S01 [1]	Low-cost (use an only camera), and generic	In-hand mobile-phone and marker placement are difficult	4.0
S02 [68]	Low-cost (use only camera)	In-hand Laptop and marker placement, not generic	2.0
S03 [69]	Low-cost (use only camera)	In-hand Laptop and marker placement	2.0
S04 [70]	Low-cost (use only camera)	Wearable PC and marker placement	4.0
S05 [71]	Low-cost (use only camera)	Wearable PC and not robust	2.5
S06 [72]	Wearable devices are easy to carry	Not extendable and Limited to blind users	2.5
S07 [73]	Wearable devices are easy to carry	Not extendable and Limited to blind users	3.5
S08 [74]	Precise and considerably accurate	Static location update	2.0
S09 [75]	Efficient, precise and scalable	Hit-rate is lower	4.0
S10 [76]	Low-cost	No audio, May fail in low light	2.0
S11 [77]	Accurate, Does not need internet	Not extendable, no audio	2.0
S12 [78]	Robust and reliable	No user ovulation	2.5
S13 [79]	Simple and accurate	Require AR tools installation	3.0
S14 [80]	Accurate being hybrid (Camera and sensors)	User has to carry both smartphone and IMU sensor	2.5
S15 [81]	Intelligent, Large scale	Difficult in installation and object selection	3.0
S16 [82]	Accurate	Only position tracking	2.5
S17 [83]	Low-cost	Marker placement is required	3.5
S18 [84]	Avoiding collision	Deformation in map	3.5
S19 [85]	Avoiding collision and Context awareness	For blinds only	3.5
S20 [86]	Reliable and Robust	Not extendable/ limited to blinds	4.0
S21 [87]	Fluent and continuous guidance	Additional cost of haptic glove/limited to blinds	3.0
S22 [88]	Accurate and robust	It requires a map pre-built step	4.0
S23 [89]	Accurate; Easy to configure/maintain	Requires Lbeacons as additional installment	3.5
S24 [90]	Efficient and minimal workload	Worse route retention	4.0
S25 [91]	Efficient and safe	Fails in collision detection of small objects	3.0
S26 [92]	Easy to hold camera	Failure with change in ground textures	2.0
S27 [93]	Efficient and robust	Failure in bright light	3.0
S28 [94]	Robust	Prior information required for training	3.5
S29 [95]	Efficient	Difficult to carry, High cost	2.5
S30 [96]	Low-cost, Robust	Limited to BVI	3.5
S31 [97]	Robust, efficient	Difficult to carry	3.0
S32 [98]	Safety, dual mode	Limited to blind	3.0
S33 [99]	Easy to implement	Manual interactions from user	2.0
S34 [100]	Combines AR and VR for accurate results	Difficult to install	2.5
S35 [101]	Accurate and efficient ML based method	May face collision, Limited to robots	2.0
S36 [102]	Comparison	Vision-based methods use laptop	3.0
S37 [103]	Robustness and generality	No user evaluation	3.5
S38 [104]	Shortest path calculation	Only 3 users in the user study	2.0
S39 [105]	Easy to use and efficient	No obstacle detection, requires QR code placement	2.5
S40 [106]	Easy to install	The tag placement is an extra cost	2.5
S41 [107]	Accurate, scalable and map-based floor plan	Difficult to install	3.5
S42 [108]	Efficient, Easy instructions	Difficult to install, cost of stereo camera	3.5
S43 [109]	Accurate	Fails to differentiate between similar images	3.5
S44 [110]	Simple and accurate (up to 1 meter)	Used only one sign	3.0
S45 [111]	Simple and with floor plan	Did not use different types of signs	2.5
S46 [112]	Low-cost	Marker placement, currently analyzed for robots	2.5
S47 [113]	Low-cost, accurate	Not robust, limited to VIP	2.5
S48 [22]	Robust, Accurate	Focused user localization only	3.5
S49 [114]	Low-cost	Requires marker placement	2.0
S50 [115]	Low-cost, efficient	Not robust	2.5
S51 [116]	Low-cost	Requires ARCore compatible machine	2.0
S52 [117]	Low-cost, easy to use	Not robust, Poor evaluation	2.5
S53 [118]	Accurate	High implementation cost	3.0
S54 [119]	Accurate and efficient	Power consumption and other cost	4.0
S55 [120]	Accurate and efficient	Cost of stereo camera, Limited to robots	3.0
S56 [121]	Double camera (robust)	Accuracy needs to be improved, Limited to robots	2.0
S57 [122]	Safe, efficient and effective	Limited to VIP	4.0
S58 [123]	Safe, efficient and effective	Limited to VIP	3.5
S59 [124]	Achieve satisfactory object detection	Fails with Dynamic change in the environment	2.5
S60 [125]	Robust and precise	No navigational guide	2.0
S61 [126]	Low-cost	Limited to VIP	2.0
S62 [127]	Low-cost, accurate	Limited to robots, Movement restrictions	2.0
S63 [128]	Low-cost, Simple	Limited to robots	2.0
S64 [129]	Robust and efficient	Sensitive to noise	2.0
S65 [130]	Accurate	No usability analysis	3.0
S66 [131]	Low-cost, accurate	No evaluation of navigation	2.0
S67 [132]	Low-cost, efficient, accurate	Maybe sensitive to light condition	2.5
S68 [133]	Robust	No usability analysis	2.5

Another hybrid method [125] is recently proposed which uses a camera with infrared detectors. The camera can utilize either natural landmarks or AR markers [126,140]. They achieved a precision of 3.5 cm in indoor positioning, however, no tests were performed for navigation. Magnago et al. [128] minimized the total number of fiducial markers used to an optimal number.

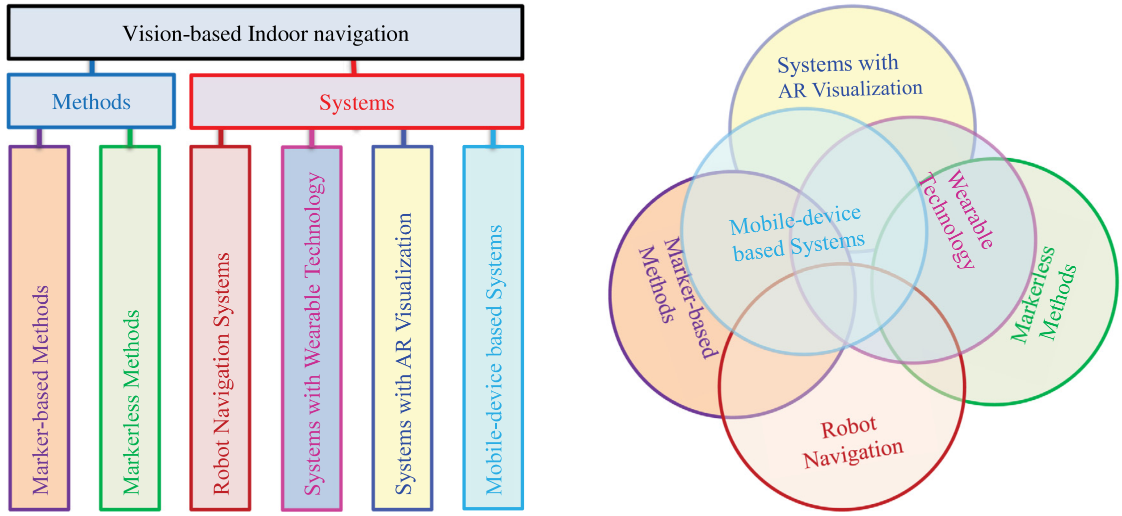

Fig. 3. Classification of the selected articles. Left: Method based and system based classification. Right: Overlaps among different categories.

They tested their system with robot navigation and found that their optimal placement ensured the accuracy with a minimal (optimal) number of landmarks.

Marker-based methods have several advantages. These methods outperform in different cases than simple image recognition and other alternative solution. A recent study [141] comprehensively analyzed three different approaches including SIFT (Scale-Invariant Feature Transform) and SURF (Speeded-Up Robust Features) and ArUco markers [140] with their advantages. The study found ArUco with better performance for indoor navigation. Similarly, another advantage of the marker-based system is its low-cost implementation. It provides robust results if we carefully design and place the markers. However, in some cases, the robustness can be affected by light intensity. For example, at higher light intensities, the light is reflected by the marker and marker detection fails. False-negative also happens with lower light intensity. Other limitations of marker-based systems include the difficulty in markers' design, printing, and placement. Ensuring marker distinction is also challenging if we need a large number of markers. The building administrator (owner) may also dislike the markers as it affects the beauty of the indoor scene being additional landmarks. The concept of hidden marker can be a future solution. Khan et al. [137] have studied 11 factors which can affect the tracking of AR markers (i.e. ARToolKit markers). Among these 11 factors, several are more important such as marker distinction, light intensity, camera quality, printer quality, and movement of the camera. They also proposed a fiducial marker optimizer [138] which is a tool used for designing distinct and high quality markers. MOMA odometry [127], which uses a mobile marker, has also set a movement restriction that at least one of the two sides (Camera or Marker) must be static. Similarly, Yang et al. [126] have summarized the following special requirements for a typical marker-based system (multiple markers):

1. All makers should be generated from the same dictionary.
2. In some cases it needs to be in the same size.
3. All markers should be distinct.
4. It needs proper light for visibility.
5. All markers should be registered in the same coordinate.

In summary, maker-based systems are easy to use and are comparatively more reliable and robust for indoor navigation. However, there are several limitations of marker-based methods such as placement of the markers, the effect of light intensity and its direction that can cause issues, and the occlusion of the markers [137].

3.2. Markerless methods

In markerless AR, no fiducial markers are placed for AR tracking. In markerless methods, the user location is traced using natural image tracking, point of interest labeling [142], through sensors, wifi, or other methods without any predefined printed patterns (i.e. fiducial markers). Similarly, hybrid systems and machine learning approaches [49,101,102] can also be used to tack the user walking trajectories to guide them toward a particular destination. There exist different AR tools used for markerless tracking. For example, ARCore [143] is an AR platform which has also been used for indoor navigation and other AR tracking systems. There exist different markerless AR indoor navigation systems. Semantic labels [142] might be logically mapped to different points of interest in the indoor environment for position tracking. In this regard, an AR-based indoor navigation (*ARBIN*) [89] is recently proposed, which uses BlueTooth low energy (BLE) beacons (i.e. Lbeacon) as a position tag to provide position coordinates and uses Google ARCore [143] for AR overlay to direct users for indoor navigation. This method starts with route finding when the user selects its destination. Then the BLE messages are used for user localizations. Directions are visually generated using AR technology. The directions are calculated by considering the expected face orientation, user motions and pitch of the IMU of the mobile phone. *ARBIN* was evaluated with different metrics described as follows. First, it was evaluated in the in-house trial for quantitative results. Android sensor manager was used for Azimuth analysis (azimuth is the angle between smartphones compass and the real magnetic north). The accuracy was evaluated and was found with 3 m to 5 m accuracy. In the first in-house trial, an accuracy of 92.5% was found. The second trial was in a hospital called field trial. ARBIN was evaluated with four users in the field trial where 35 Lbeacons were deployed. The field trial showed that 97% of the users' feedback were moderate. As future work, ARBIN can be extended to add landmarks to the real world to enrich the user experience.

Similarly, another vision-based navigation (*ViNav*) method [75] used 3D models and navigation meshes for indoor navigation. *ViNav* used a hybrid approach of visual and internal sensor data from smartphones. *ViNav* used structure-from-motion (SfM) for 3D reconstruction of indoor scenes from crowd source visual imaging data and points of interest (POI) identification in 3D models of the indoor environment. The system was evaluated in two different places via 13 participants. Different metrics were noted during the experiments including hit-rate, accuracy (location error and direction error), robustness, latency, and scalability. *ViNav* is a cost-effective and efficient system for indoor navigation. It also supports multi-floor mechanism. It can locate users within 2 s. The localization error is smaller than 1 meter and the face direction error is less than 6°. However, the hit-rate was somehow not good like WiFi and image-based systems.

Kanwal et al. [93] another method which uses camera and infrared sensors for depth perception. With a focus on visually impaired people, this method is based on object detections which is achieved via corner estimation. Furthermore, depth perception is achieved using Microsoft Kinect camera. The authors found this method being efficient and robust in normal lights. However, infrared sensors fail in depth perception in bright lights. Furthermore, the small obstacles are often not detected. Moreover, it is difficult to set up hardware (Kinect Camera, touch screen and CPU) during navigation. Another system uses an RGB-D camera for depth perception [96] which provides obstacle-free paths for safe navigation. The system was escalated in various datasets and environments. The system was found robust with different light conditions. It was found with 99% accuracy and 95% of recall.

Another markerless method [97] tracks different landmarks (e.g. stairs, fire extinguishers, windows etc.) and combines them with existing geographic information system (GIS) data for location tracking and navigation. The system works in real time and can find a path from source to destination. However, it is currently processed via computer which is difficult to carry. Therefore, a smartphone implementation is a future direction. Similarly, a single camera with 3D map [131] is another simple idea for accurate positioning. Though there is no specific evaluation on navigation, this study [131] achieves an accurate positioning with an average error of 10 mm. Co-Robotic Cane (CRC) [98] is another navigation system for blinds and visually impaired people. CRC uses a 3D camera for pose estimation and object recognition. As it does not need prior knowledge of the environment, it can work with unknown environments. CRC works in two modes: in active mode, it can direct the user by directing itself toward its destination; whereas in passive mode, it is a computer vision assisted white cane. This dual mode increases its safety as failure in robotics functionality will automatically convert it to conventional cane.

Guerrero et al. [124] used two Infrared Cameras (embedded in two different Wiimote devices) for user tracking and a smartphone for receiving navigation guides (in audio). They mainly focus on usability enhancement and achieve considerable improvement. However, their map estimation fails with dynamic change in the environment (such as changing the position of a table). Another method [130] used an RGB-D camera (mounted on the user's forehead) for environment sensing and creates a point cloud map via SLAM technology. This is an AR-based indoor navigation system developed for smart glasses and can be used as a tourist guide. Furthermore, an orientation error correction approach is used to improve the accuracy. The experimental results revealed that the system achieved 99% accuracy. However, the lack of usability analysis makes the reader doubtful about user satisfaction such as ease of use and fatigue during its usage.

Tian et al. [103] proposed another navigation method which uses the computer vision to recognize text, different regions of interests, walls, and door images to ensure user localization and tracking. The method was specially developed for blind users; therefore, it takes the destination from users as an audio input. Similarly, it provides audio output to the users. The system was tested with different image datasets and was found to be robust. However, there are no user evaluations. Furthermore, though it can be generic, their study was limited to blind users. Another method uses the $D*$ algorithm for shortest path calculation and the Continuously Adaptive Mean-Shift (CAMShift) algorithm [104] for object recognition. However, the system was not properly evaluated to strengthen their finding. They used only 3 blind users for blind navigation. Markerless systems are relatively easy to install as there is no burden of handling markers. However, finding distinct objects or images for user localization is challenging due to the high similarity between indoor images.

4. Indoor navigation systems

In this section, we presented state of the art indoor navigation systems in four different categories including robot navigation based systems, wearable technology based systems, systems with AR visualization, and mobile device based systems. A number of articles have already been presented in Section 3 in two categories including marker-based and markerless methods.

4.1. Robot navigation systems

Typically, robot navigation systems and blind navigational aids have similar mechanism. In this regard, a recent study [101] has used deep a convolutional neural network *(DCNN)* for robot indoor navigation, which can also be generalized for general navigation of the people. Moreover, collision detection and avoidance can be interesting ideas. Similarly, Botta and Quaglia [112] used *ArUco* markers [140] for mobile robots. However, their system can be used for general indoor navigation. Cheng et al. [120] used inertial measurement units *(IMU)* of the mobile robots with stereo cameras for indoor localization. They used dual Kalman filter (DKF) algorithm to improve accuracy of the system. However, their current implementation was only evaluated with robots.

Diop et al. [121] implemented robot navigation system through two cameras on a mobile robot. Their system gave arrow shape passive beacon to robots as a navigation guide. This system can be further improved for human navigation in indoor environment. MOMA [127] used mobile markers for position tracking and robot navigation. Experimental results showed that MOMA has an error of 0.045 meters i.e. 0.138 percent of the total path. However, there is no experiment for human navigation. Moreover, the camera should be fixed as the marker is in movement. QRPos [129] is a QR based system with an optimized scanning algorithm. It is a robust, efficient, and accurate system; however it is susceptible to noise. Robot navigation in indoor scene also includes the key functionalities like human navigation system. These functionalities include building a map, path planning, positioning, and collision detection and avoidance, and tracking robot's navigation trajectory [144]. These methods can directly be extended for human navigation especially for blind users. Perhaps the feedback from a robotic system might be somehow different. For example, the robotic signal will need a translation to audio or visual information for human navigation. Similarly, the robot navigation methods are generally evaluated with few robots (mostly one robot) with main focus on localization errors. Since robotics methods are completely autonomous, they are more challenging than user navigation. Environment understanding and learning spatial knowledge can play a vital role in robot navigation [145].

Although, robot navigation systems can provide a baseline to human navigation. Still, there are several limitations with robot navigation. For example, they typically need an expert to install, consume power, and are fully dependent on predefined commands. There are different robot navigation methods which use inertial sensors and signal processing. However, vision-based methods are comparatively lesser than other alternatives as robots are mostly found with strong sensors which may work more robustly than a camera. After minor extensions, the robot navigation systems can be utilized for human navigation.

4.2. Systems with wearable technology

Wearable devices are in two categories including vision-based methods and others. Our focus is on wearable vision-based technologies. Headloc [72] is a navigational aid developed for a blind user that detects landmarks using a wearable Google Glass. Audio feedback is generated after landmark detection. Headloc was evaluated with eight blind users. Text-to-speech and audio feedback were analyzed in the user study and text-to-speech was preferred by the users. The evaluation was carried out using qualitative and quantitative metrics and significant results were found. However, there is no experimental study to compare Headloc with a camera. Also, there are no experiments on the maximal distance between landmark and Google Glass. Ebsar [73] is another indoor navigation system for blinds. It also uses Google glasses connected to a smartphone. It uses QR code markers for the position of interest. Smartphone accelerometers and compass sensors were used to determine the distance and direction between the QR codes. Ebsar was evaluated via 22 users in three different metrics including effectiveness, efficiency, and user satisfaction. The results were found satisfactory, however, Ebsar is limited to blind users.

Rehman and Cao [74] have developed another Google glass-based indoor navigation system. They have used a 3D point cloud for environment tracking. They evaluated their system with different metrics such as accuracy, time, and feature tracking. Furthermore, they have evaluated it via a user study of 27 participants. The system was found precise, efficient, and user-friendly being with minimal cognitive load. However, there was no dynamic update which may cause a delay in location updates. Wearable devices in general are easy to carry while navigating in indoor navigation. Furthermore, the tracking technology has also very suitable for indoor navigation. However, these techniques are with several limitations such as dependency on short battery life, slow processing, installation cost, and low-quality visual results. A recent method [122] used wearable optical see-through glasses which utilize ultrasonic sensors and RGB-D Cameras for object recognition and obstacle avoidance. The system [122] was found effective, safe (with collision avoidance), and efficient. A similar previous work [123] used smart eyeglass, RGBD Camera, and ultrasonic sensors. However, it is limited for VIP and the implementation cost is also high. Recently, the system has been further extended [91] with additional modules.

Another wearable system [94] used an RGB-D camera, an IMU, and a laser sensor for obstacle avoidance, and helping blind people with indoor navigation (see Fig. 4). It uses Aruco markers as landmarks in the indoor environment. Due to multiple sensors, the method is robust and efficient. It emphasizes both recognition and navigation. However, this system has a high implementation cost and it is difficult to carry during navigation. Furthermore, it is limited to blind users, so a generic system for general indoor navigation can be a future extension. RGB-D camera [84] has also been used as a wearable navigation system. In addition to RGB-D camera, the system [84] used different sensors (see Fig. 5) including IMU, smartphone, and tactile feedback. The system was evaluated with six blind and four more blind-folded users. The system was found with significant improvement in indoor navigation and avoiding collision. However, there are a few limitations of this system, for example, inconsistency in the map and low accuracy. Furthermore, it was specially designed for blind users. Moreover, it is difficult to wear all the sensors used as shown in Fig. 5.

Intelligent situation awareness and navigation aid (ISANA) [85] is another wearable navigation system for guiding blind people through audio instructions. ISANA includes semantic maps for context-aware navigation and an efficient algorithm for collision detection and its avoidance. Dynamic modeling for future

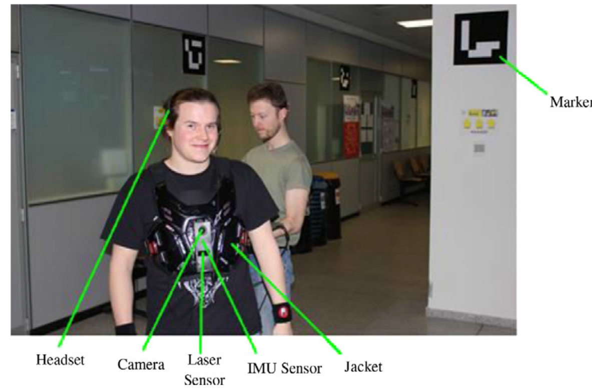

Fig. 4. Aruco markers as landmarks in the indoor environment and different tracking sensors including laser, IMU and camera [94].

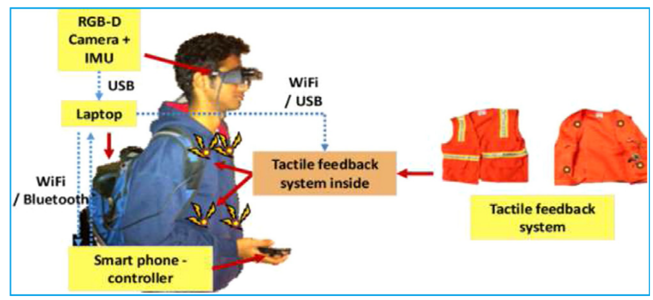

Fig. 5. A wearable navigation system using a head-mounted camera, IMU sensor, smartphone controller and tactile feedback [84].

collision detection and environment understanding are the future directions as highlighted by the authors. ISANA was further improved with semantic maps from 3D models and a CCNY SmartCane for the robust human–machine interface at the City College, the City University of New York (CCNY) (see Fig. 6 (Left)) [86]. The results showed a significant improvement compared to the previous version of the ISANA. However, there are still further challenges for the future such as robustness in noisy environments, an easy-to-use interface for adjusting audio frequency, adding landmarks to the map. Furthermore, it is limited to blind users so it can be extended to a generic system for implementation in large and complex indoor environments. The authors also intend for cognitive understanding as future work.

Bai et al. [91] proposed a wearable technology (eyeglasses) and smartphones (see Fig. 6 (Right)). The technology used in it includes an RGB-D camera and an Inertial Measurement Unit (IMU). The system allows quick and safe navigation of the visually impaired in an unfamiliar indoor environment. The system was evaluated by 20 participants and was found safe and efficient. However, the collision for small size objects is failed sometimes.

Blind guide [92] is another system that uses a video camera as a virtual eye for indoor and outdoor navigation. It uses a stereo camera for location tracking and generates audio information to guide the users. It avoids collision and generates path from source to destination. The camera is mounted on the user's head, so it is easy to carry compared to handheld devices. However, it may fail for an abrupt change in the ground or the texture of the ground. Another wearable system for indoor navigation of blind or visually impaired (BVI) is achieved via camera, embedded computer, and haptic feedback [95]. This system [95] helps the BVI in indoor navigation by guiding them to free space, avoiding collision and path to a specific destination. In the subjective analysis using 15 participants, the system was found with low latency feedback and

Fig. 6. Pipelines of the two existing technologies for indoor navigation. Left: ISANA and CCNY SmartCane for a blindfolded participant [86] *(Copyright © 2018, IEEE)*. Right: RGB-D camera and IMU based wearable technology (eye glasses) for indoor navigation [91].

with high frame rate. Energy-efficient implementation is planned as a future work by the authors. A stereo vision-based system [108] was recently proposed, which uses a stereo camera, implements floor segmentation, and avoids obstacles to help blind users for a safely wander in unknown environments. The stereo camera is placed on the head or chest as a wearable system. The system [108] was tested with 10 participants and found efficient and acceptable to users. However, the stereo camera is an additional cost and the installation is also difficult.

Generally, wearable technology provides a more realistic framework for indoor navigation as it can give more feedback to the users. It gives haptic feedback in addition to other types of feedback such as audio and visual. Furthermore, wearable devices are easy to navigate with as compared to handheld devices. However, the cost is relatively higher, and the installation is difficult for these systems. Similarly, wearable systems are not typically stand-alone. It can cause some health risks and the user may feel socially uncomfortable. Continued power requirement is another issue with wearable systems.

4.3. Systems with AR as visualization only

In this section, we present indoor navigation systems, which use some other methods for position tracking but use AR technology for visualizing the navigation information. *INSAR* [77] is an indoor navigation system with augmented reality. *INSAR* used WiFi fingerprinting for position tracking and AR for visualization of the navigation information. Similarly, a digital compass is used for direction information. The system is evaluated by seven participants and was found with significant accuracy. It does not need an Internet connection. However, it has several limitations such as not being extendable, and cannot be applicable for any change in the environment i.e. changing the access points. Furthermore, there is no audio guidance for the user. Chandgadkar [146] used custom 2D colored markers detected via smartphone for indoor navigation. He used an accelerometer for step detection. However, there was no mechanism for automatic path planning. Furthermore, the system fails in low light as the colored markers are miss detected. Furthermore, it was not extendable to multiple floor buildings.

STEPS [79] is another vision-based indoor navigation that uses mobile phones. STEPS uses AR technology using Google ARCore and Apple's ARKit. It is a simple, efficient, and relatively accurate method for indoor navigation. ANSVIP [87] is another indoor navigation system for visually impaired people. ANSVIP also has a path planning algorithm that avoids possible collision. Haptic interactions are used human-centric efficient motion delivery rather than state-of-the-art turn by turn wayfinding schemes. ARCore was used for the implementation of various modules including pose tracking, map building, and area of interested detection using smartphones. ANSVIP was evaluated with visually impaired and blind participants and was found with significant achievements.

Rehman and Cao proposed another AR-based navigation [90]. They carried out a subjective analysis on 39 participants. They compared their system using handheld devices and Google glasses; and found that wearable devices are more accurate, however, other performance and workload results have no significant difference from handheld smartphones. They found that both of the implementations are more efficient and effective than a paper map, however, both had worse route retention. The compatibility issue of ARCore is another limitation. Möller et al. [100] proposed a mixed reality interface that uses virtual reality (VR) and augmented reality (AR) elements as indicators. These indicators are used for communication and accurate localization. The system was evaluated with 81 users. The result revealed that AR is better in reliable localization while VR was found better for accuracy and navigation instructions. Typically, AR visualization easily provides more details. AR visualization is also found in most of the methods in marker-based and markerless methods.

In usability concerns, the AR-based systems are the optimal solutions. AR visualization can give the user an interactive framework where the user can explore a particular destination and find an optimal path toward a destination. AR visualization can give more confidence to the navigating user as the user can see the valid and optimal path and its dynamic updates with user navigation and time. The main issues of AR visualization are cognitive load and partial cut off from the real world. In other words, the user may feel tired or in burden while looking toward AR visualization and also navigation in the real environment. These systems can be extended to a purely virtual environment to be used for a remote tour.

4.4. Mobile device or smartphone-based systems

Though a majority of the aforementioned articles have used smartphones. In this subsection, we describe a few articles which use the camera of smartphones as well as other features of the smartphones such as NFC tag reading [78] or accelerometer [73]. Mobile devices have different sensors which can help the mobile camera as additional support to form a hybrid navigation system (see Fig. 7). Wu et al. [82] have compared the human eye with smartphones' camera. They developed a visual system where the position is calculated from a single image using a smartphone camera. They found that the performance of the smartphone is more accurate than human brain perception according to their experimental setup. This method can improve the human perception of their location in indoor environments. Indoor image features are also an alternative to markers which are used for location tracking and indoor navigation [117]. This interior feature-based system [117] was evaluated in a shopping mall and was found user-friendly. However, the evaluation method is not well analyzed. These features might be falsely detected if the light intensity is not optimal.

Labelee [78] is an indoor navigation system, which scans QR codes and reads NCF tags for location tracking. The authors plan to use smartphone compasses to show direction toward the user's destination. Being the use of wireless technology, Labellee is expected to be robust and reliable in any indoor environment. However, the user evaluation of the system has not been carried out so that the study claim can be proved.

A recent hybrid indoor navigation system [80] has used a smartphone camera in conjunction with an inertial measurement unit (IMU). This hybrid system was statistically evaluated and was found the significant accuracy. They used ArUco markers [10,140,147] tracked by camera. ArUco markers were also used by another multisensory indoor navigation system [148], with a

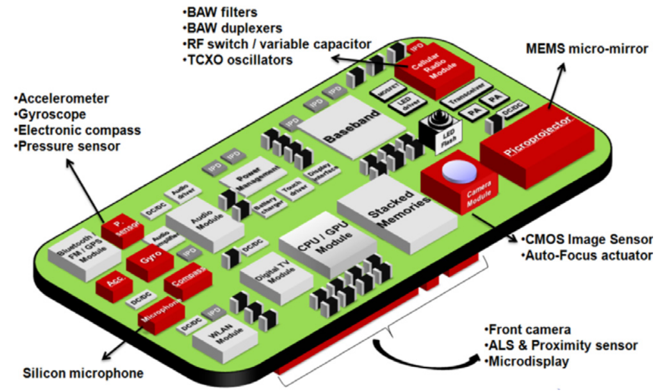

Fig. 7. Mobile internal sensors used for indoor navigation [31].

specific focus on micro air vehicles. The use of a hybrid system (camera and IMU sensor) [80] has improved accuracy, however, it is difficult to carry both smartphones for recording via camera and IMU sensors. The authors intend to further improve the accuracy of this hybrid navigation system [80] by using optical flow and ultrasonic sensors in the future.

Xiao et al. [81] proposed an indoor positioning system based on static objects recognition using the smartphone camera. Their objective was to deal with relatively large indoor environments. They detected static objects placed in the indoor environment which is then used for location tracking. For object recognition, they proposed an intelligent recognition system by using deep learning and computer vision algorithms. It achieves accuracy within 1 m in a range of 40 m of the indoor environment. However, the initial installation such as static object selection is difficult and needs special care. For example, the objects should be evenly distributed, not too small, easy to capture images, and to a degree distinct from each other.

Vision-integrated fingerprint [88] is another mechanism that uses Wifi with a vision-integrated fingerprint to achieve accurate and robust position tracking in an indoor environment. Their experimental results showed 95% to 98% rate of site recognition, and the average localization error was less than 0.5 m. However, it requires the environment information in a preprocessing step; so it does not apply to the new environment. Similarly, another method [132] used a video camera to detect user walking speed via a smartphone. This method gave map-based indoor navigation for pedestrians as well as wheelchairs users. The evaluation result by 11 participants raveled that system is accurate and efficient. Visual odometry [133] is another system that used a smartphone's camera to estimate user trajectory using visual SLAM. Salient image points are used for depth estimation. An Extended Kalman Filter was used for estimation of the scale factor to metric coordinates using inertial and visual information. The system was evaluated with an indoor dataset (i.e. *TUMindoor*) [149] and was found with satisfactory results. Smartphone-based indoor navigation systems are easy to install and implement at a low cost. However, holding the smartphone in the right position is difficult during navigation. It makes the human hand tired [90]. Idrees et al. [105] used QR code detected via smartphone's camera. Their system is efficient, to a degree accurate, and easy to use. However, they have no obstacle detection. Furthermore, their system is limited for blind users and does not look realistic. In addition, the installation of QR codes in the building is an additional task.

Perera et al. [115] proposed indoor navigation for complex shopping malls where they used CNN for detection of shops signboards. They evaluated their system with ten users and found

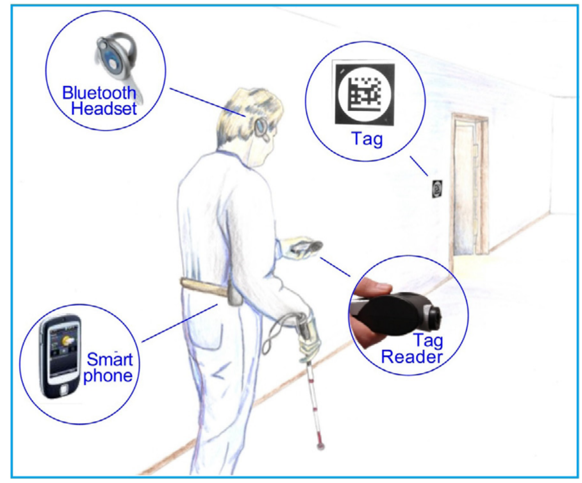

Fig. 8. Indoor navigation using a digital signature system [106].

it accurate and low-cost. ARCore [143] has also been used for smartphone-based indoor navigation system [116]. This system [116] detects different images for location tracking and provides a visual map for indoor navigation. The system was evaluated with 15 users using the SUS questionnaire and was found with significant user satisfaction. However, the image recognition is not robust with different light conditions. Moreover, ARCore has compatibility issues and different mobile devices are not compatible with ARCore. Other previous systems such as ANSVIP [87], ARBIN [89], and STEPS [79] have also used ARCore. Shahjalal et al. [118] used multiple LED transmitters giving location information to a mobile device using LED lights. The system was found accurate; however, the implementation was difficult and expensive.

Travi-Navi [119] is vision-based indoor navigation that uses high-quality images for indoor navigation. In addition to these images, Travi-Navi also uses magnetic field distortion and Wifi sensors to improve the accuracy of user tracking. Though the system has been evaluated with only four users with no specific subjective assessment (qualitatively). Still, the experiments were conducted on different paths and different aspects (i.e. efficiency, accuracy, energy consumption, and user tracking) of the system have been analyzed very well. Galioto et al. [113] used internal sensors of the smartphone and achieved a significant accuracy in indoor navigation. However, their system is not robust with different light intensities. Similarly, Zhao et al. [22] used Wifi signals with smartphones camera for indoor localization. They found their system with 92-percentile error was with 0.2 meters from their points of interest. Since the system uses two different technologies (wifi+camera), therefore it was found robust. However, the study focus is only on user localization and there are no specific guidelines for indoor navigation. Furthermore, there is no obstacle detection and floor plan.

Digital signature [106] is another smartphone-based system used for indoor navigation, which works on digital tag detection (see Fig. 8). However, tag installation its reader are additional costs that make it difficult to install than other alternative solutions. Similarly, text recognition using a smartphone camera is another method for indoor navigation [99]. The landmarks are detected via a smartphone camera and the text over it is sent to a database at the back end which is used for user pose estimation and pathfinding. However, the user in this method [99] has to manually decide about the selection/rejection of the detected landmarks. Vivid [107] is another mobile system that augments vision-based navigation with edge computation in an indoor environment. It has used the deep learning method for dealing with variable light conditions. Furthermore, Vived used a grid-map-based algorithm for the floor plan. The experimental results showed its accuracy and scalability. However, the installation is difficult as the system has used several algorithms.

CamNav [109] is another navigation system that uses multi-scale local binary pattern (MSLBP) features [150,151] for scene recognition. It is a client–server system that has a neural network model, communication and processing logic, and dataset on the server side whereas a mobile device on the client side. The system has achieved an accuracy of 97.02% which is higher than oriented FAST and rotated BRIEF (ORB) [152] accuracy (which was 96.33%). However, CamNav often fails to differentiate between similar scene images. Furthermore, the authors provide no usability analysis. Another smartphone-based method [110] implemented user localization via exit sign recognition and internal sensors of the smartphones. The method was further improved [111] with floor plan and dead reckoning for user movement estimation. However, the method is limited to exit signs only and it needs to be extended for different signs such as stairs, restrooms, and room numbers. Furthermore, the experimental evaluation is also not well described.

Smartphone-based systems are easily available at a lower cost as smartphones are generally available to most people. In addition to the camera, a number of internal sensors available in smartphones (see Fig. 7) have been utilized with camera (hybrid system) as well as without camera. In summary, smartphone-based systems are low-cost, easy to use and there is no need for any technical expertise. However, there are a few issues with these systems. The main limitation of the mobile device-based system is to carry it during navigation. The user has to turn on the camera during navigation so that it can recognize the target images, objects, or markers. A typical method is a handheld mobile device that is not convenient for the user. Compatibility among different mobile devices is another issue of this category. Smartphone-based methods are overlapping with other categories of methods (see Fig. 3).

5. Evaluations metrics and data for indoor navigation

One of the objectives of this study is to help new researchers in the field. Results analysis is one of the main parts of the research. Therefore, in this section, we briefly present the way the state-of-the-art indoor navigation methods have been evaluated. There are two major types of analysis including quantitative and qualitative results. The quantitative analyses include the accuracy navigation, accuracy of marker/image recognition, collision detection, and false positive and false negative of the landmarks (placed at different points of interest). The qualitative results are the subjective analysis of the system through user study and questionnaires; used for usability testing and user satisfaction. Similarly, the ease of installation and the installation cost are also important factors.

5.1. Analysis criterion for indoor navigation research

Typically, indoor navigation systems/algorithms are analyzed either with subjective analysis (user study) or through some quantitative metrics such as efficiency and accuracy. The subjective analysis is carried out via a simple questionnaire or by following some standard format such as system usability scale (SUS) [153] are also used [1,116].

A user study is the most common method for evaluations. However, finding a suitable number of users can be a challenge. Especially, for blind users, it becomes further challenging. For example, in one study [104], the user study was carried out with only three users. An alternate solution to this issue is to involve blindfolded users. The user study usually focuses on usability study and user satisfaction. However, different quantitative data can also be collected during the experiments, e.g., false detections, miss detections, accuracy, and time-consuming. For the usability analysis, in addition to SUS, other AR usability concerns [154] can also be used.

The lack of a standard evaluation method is a key issue in indoor navigation research [155]. Few metrics have been recently collected from the International Standards Committee of IPIN. These metrics include floor detections, point of interest detections, statistical errors, latency, start-up time, availability, efficiency, type of building, and mobility [155]. Though this study [155] is general for all indoor localization, still it is important for vision-based indoor navigation. In addition, we can consider other factors such as user satisfaction and usability analysis, implementation cost, and time comparison. Adler et al. [156] presented different evaluation criteria for indoor localization adopted in *IPIN* conference from 2010 to 2014. They collected different metrics used for the evaluation of indoor localization including trueness, accuracy, precision, distribution, and sample size. They found that 95% of the articles have evaluated their systems (with simulation or physical experiments). They found that physical experiments are found with a higher ratio i.e. 77%. In the context of simulation, the trend toward complex simulation is higher than the simple one. They categorized the articles based on their evaluations as follows.

1. **No Evaluations:** Some articles have neither been evaluated with simulation nor with experiments.
2. **Simulation:** Some articles conducted **simple simulation** with no errors and obstacle considerations. Other articles conducted **Complex simulation** considering different types of errors and obstacles during simulations.
3. **Discrete Point:** Experiment on a few selected points of interest.
4. **Grid-Like:** Experiments in an empty indoor environment with no obstacles in it.
5. **Office Walk:** It considers a specific path in the indoor environment. The user is moved on the path and compared with the ground truth.
6. **Real World:** The experiments or simulations are performed on realistic indoor environments with population and all real-world challenges such as obstacle avoidance.
7. **Multiple:** Some articles used more than one evaluation criteria.
8. **Unrelated:** Some articles have different experimental results not related to user localization.

Consecutive Stability is another metric used in speed detection based method [132]. It refers to the consistency of algorithm updates with user actual movement. Evaluation with some existing datasets has also been found in the literature [133,149]. Zheng et al. [119] have used multiple quantitative metrics in their experimental study including time, power consumption, motion tracking, shortest pathfinding, localization error, and detection of the crossing points (T and $+$ crossing points). Localization errors and distance from the actual point of interest are the significant metrics to be used for experimental analysis [22]. Room mean square (RMS) errors [120] is another quantitative metric used for experimental evaluations in indoor navigation. RMS error is the RMS distance between the actual and estimated location.

The general requirements for the indoor navigation systems are implementation cost, power consumption, memory consumption, ease of maintenance, and ease to be updated [25].

5.2. Datasets used for indoor navigation

From the literature review, we identified two broad categories of indoor navigation methods including marker-based and markerless methods. Markerless methods are typically founded with some general object recognition or the utilization of modern machine learning algorithms. Therefore, in addition to subjective analysis, different authors have used public datasets for analysis of their algorithm/system. In this section, we provide a short overview of a few well-known datasets publicly available to the indoor navigation community. For example, Teng et al. [157] have used three different types of datasets. Oxford Step Counter dataset [158,159]. Aladren et al. [96] have used special dataset (http://webdiis.unizar.es/~glopez/dataset.html) for their results analysis. They used different statistic parameters including precision, recall (calculated as the true positives divided by the total), and F1 statistic according to the segmented area of the floor.

Robot navigation is also typically aimed with existing datasets. In this regard, social robot indoor navigation (SRIN) [160] is a dataset of 2D images containing *37,288* images of doors in five classes (including bathroom, kitchens room, general bedrooms, and dining) and *21,947* images for doors in three states (open, close, no-door). (RISEdb) [161] is another indoor localization dataset. The dataset contains more than 1 million images captured from 20.7 km long which is a 6 h walking path. This dataset provides a large set of labeled data to train any machine learning algorithm for indoor navigation. Navigationnet [162] is another similar deep reinforcement learning dataset. Indoor Robotics Stereo (IRS) dataset [163] can also be used by deep learning-based algorithms with a special focus on surface normal estimation. It contains 0.1 million stereo RGB images, maps, and calculated surface normals.

MIT dataset for indoor scene recognition [164] contains 15620 images in 67 different classes. *GTA* indoor motion dataset *(GTA-IM)* [165] is concern with user interaction in indoor environment. Another dataset [166] which contains images captured with a smartphone camera and *LiDAR* scanner can be used for indoor localization. This dataset can be used for quantitative analysis of the indoor navigation systems/algorithms. Autonomous robot indoor dataset *(ARID)* [167] is another public dataset with more than *6000 RGB-D* images and *120,000 2D* bounding box of various objects. *ARID* can help robots in different applications including indoor navigation and object detection. Object clutter indoor dataset *OCID* [168] was created via EasyLabel which consists of RGB-D images and labeled point-could of different objects. It can be used for indoor localization, floor plan, and object recognition.

Recently, a new dataset called *InLoc* (indoor localization) [169] was created by capturing indoor images at different times using mobile phones, to provide a realistic scene. The key focus of *InLoc* was on *6-DOF* poses for large scale indoor localization. *D-HAZY* [170] and other depth perception datasets [171,172] are also used for indoor localization and obstacle avoidance. D-HAZY [170] is specifically focused on quantitative analysis of the dehazing algorithms [173]. *SceneNet* [174] is another public dataset with several labeled images of indoor scenes in different categories. Wireframe [175] is another collection and public dataset containing 5462 images (with 5000 training images and 462 test images). For fusion of multiple sensor and depth perception, there are different datasets including *RGB-D Dataset 7-Scenes* [176], Imperial College London (ICL-NUIM) dataset [177] and The TUM Stereo Visual-Inertial Event *TUM-VIE* dataset [178].

In summary, there are different types of publicly available datasets for the use of researchers/users in indoor navigation. These datasets include users steps counting [158,159], user tacking [165], robot aids [160,167], machine learning datasets [161,162,174] and general object recognition or obstacle avoidance [168,170]. The images used in these datasets can be either 2D or with depth information or a combination of both. In addition to the aforementioned datasets, some general datasets such as *PanoContext* [179,180], and active vision dataset *(AVD)* [181] can also be utilized for indoor navigation or obstacle avoidance.

6. Future directions and challenges

6.1. Improving portability and mobility

The main evaluation method is usability analysis and user satisfaction. In this context, it is desired to have a system that is easy to carry during navigation. Though several systems have used smartphones and small devices. Still, there are different systems [97] that use computers for computation. So, smartphone implementation can be future work. Furthermore, carrying a smartphone in hand is difficult; therefore, improving the camera freedom to track the target landmarks in a wide range is also an interesting direction for future research.

6.2. The use of image based localization

Several interested directions have been mentioned by Lee and Medioni [84]. For example, image based location tracking can improve efficiency and robustness. Furthermore, there is inconsistency and deformation in the map [84], which can be considered as future work.

6.3. Generic indoor navigation systems

There are several indoor navigation systems designed especially for visually impaired [84] can be generalized for the use of common people as well. For example, the feedback for visually impaired people is usually in audio, so more visual feedback can be added for common people.

6.4. Machine learning for indoor navigation systems

In order to improve the performance from the previous experience, different machine learning algorithms can be used. It will help to improve the accuracy and find more optimal paths based on experience. In other words, these systems can learn from the daily usage of the users and their feedback. The gained learning can be later used for optimal path planning. Furthermore, the system can get feedback from the users and can update itself accordingly. Though there are few methods with machine learning [49,101,102], however, they are not very effective.

6.5. Environment understanding

Most of the existing methods [1,88] require the environment information as pre-processing step; thereby they are not directly applicable in an unfamiliar indoor environment. It is an interesting idea to introduce an indoor navigation system, which does not need a pre-map building of the environment. Environment understanding can also help in path planning and predicting the future collision is another interesting direction that can improve existing methods such as ISANA [85]. The cognitive understanding of the environment and extendibility of the system are also challenging and important for future research [86]. The abrupt change in the indoor environment will not affect if the navigation system takes some additional care such as choosing some fixed features (those not affected abruptly).

6.6. Hidden markers

One limitation with marker-based indoor navigation [1] is the placement of markers in the building; which is usually observed as a negative impact on the building beautification and decoration. Therefore, if we use some hidden markers such as existing signboards, doors, and windows or indoor scene images; it can be a possible solution to the issue.

6.7. Image to 3D building conversion

One possible direction is the construction of a 3D virtual environment from 2D images of the indoor scene. The virtual environment can be accessed remotely or in the building, and the user can navigate in it to explore different points of interest. This includes realistic 3D reconstruction of the virtual environment from 2D images. Like the official website of an organization, it can give 3D access to the remote user to virtually navigate in the organization for information gain, pre-visit, or even a virtual tour.

6.8. The use of other sensors with computer vision

Multiple sensors can improve efficiency and accuracy. Therefore, if we use additional sensors with a computer camera, the system performance can be significantly improved. For example, with marker recognition, if we utilize wifi signals or internal sensors of the mobile device, we can minimize the possible miss detections of the markers.

6.9. The use of multiple cameras

If we use more than one camera, the accuracy and the efficiency can be improved. The utilization of CCTV cameras (which are usually installed in different buildings) can be a possible solution. Multiple cameras can handle the issue of occlusion and can help in detecting collision detection. Shahjalal et al. [118] have used multiple fixed LEDs for generating LED light giving location information to a mobile device. Similarly, fixed cameras [182] and stereo cameras [183] have also been used. However, the use of existing cameras (such as security cameras) as additional support to indoor navigation is still challenging. In addition to their own main function (such as security), these cameras can be used to help in indoor navigation. Furthermore, unlike LED transmission [118], these cameras can work on simple human detection or simple feature analysis to support indoor navigation. Similarly, a user may also hold more than one camera to increase the robustness of the computer vision-based indoor navigation system. Similarly, more than one mobile camera can also be used to improve robustness. Diop et al. [121] have used two mobile cameras; however, their system is limited to robots only.

6.10. The use of language translations

There are many systems based on single systems so language translation or the use of multiple languages can be a small yet valuable contribution. Existing translator such as google translator can be utilized. Alternatively, a different option can be given to the user to select the local language and provide interface and feedback in these particular languages.

6.11. Map data availability

Though location tracking is a challenging task, still, there are enough techniques available to handle it. However, map generation in a dynamic way is still challenging. One solution can be the availability of map data such as in downloadable form from the web as suggested by Idrees et al. [105]. The extension of the existing map such as adding a new floor can be an additional option.

6.12. Extension for outdoor navigation

Different indoor navigation systems can be extended for outdoor navigation with additional inputs from the outdoor environment. Navigation in the close boundary regions such as parks, museums, zoos, etc., can be an easy extension. Similarly, some novel ideas from outdoor navigation [184] can also be adopted in indoor navigation.

6.13. Simultaneous localization and mapping

Indoor image recognition is a challenging task and the existing systems often get confusion to differentiate similar images [109]. Simultaneous localization and mapping can handle this issue. The walking history of the user and environmental context can also play an important role to handle this issue. The use of machine learning approaches can help to implement these suggested methods. Another alternative is to use image database [185] for adding semantics to environment and generating maps.

6.14. Depth images for indoor semantics

Inspired by indoor scene generation from 3D images [186], a more semantic navigation system can be designed. Unlike scene generation [186], the 3D semantic images can provide an intelligent understanding of the indoor environment, and thereby it can give a route toward a given destination from any indoor source position. Furthermore, users can virtually explore the 3D virtual environment before his/her actual physical visit. Though implementation of such a 3D environment can be challenging, still it will provide a realistic experience to the users. The new user can get experience in a virtual environment imitating any unknown environment. The time-of-flight 3D imaging [187] can also be utilized for indoor navigation. Similarly, RGB-D cameras [184] can also be adopted.

6.15. Local map sharing between mobile users

An active communication of the navigating user with other users (such as a user sitting at the source or an experienced remote user) may help in indoor navigation. Similarly, map sharing and message passing can also be used as additional guides for navigation [132]. In other words, the end users can be provided an option to communicate with each other during navigation. Alternatively, the user can ask for different information which will be provided by the system administrator or an intelligent virtual agent.

6.16. Voronoi diagrams and mesh network

Inspired by the point cloud approach [19], computational geometry can be adopted in future research. For example, the use of Voronoi Diagrams [188,189] and 3D meshes [190] for indoor localization as well as path planning is also a future challenge. The geometric understanding of the indoor environment can play a significant role in accuracy and efficiency. It can provide an easy interface for visual information and directional instruction in indoor environments. The data structure of 3D meshes [190] can provide an interesting setup to organize a different point of interest in any indoor navigation. The vertices in a surface mesh can represent different points of interest and the edges can be the path among different locations. Similarly, the installation can also become easier as it will be a simple mesh generation. For example, we can start a video camera with some input to highlight different points of interest. The video recording will generate a 3D mesh having connectivity and sequence information of different vertices. One major advantage of this data structure will be

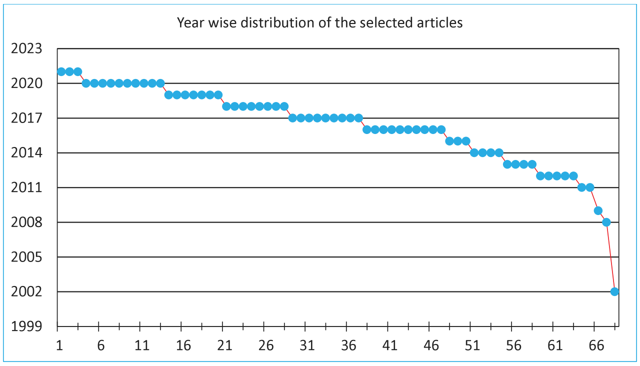

Fig. 9. Distribution of the articles on the basis of publication year.

efficiency. For example, if a user visited vertex v, then for the next vertex recognition the algorithm will search it only in the vertices list of directly adjacent vertices of v instead of all mesh. The use of Voronoi Diagrams [188,189] and other intelligent geometric algorithms/data structures can lead to further effectiveness of the system.

7. Discussion and conclusion

Indoor environments have limited access to the GPS. Therefore, location tracking and navigation in the indoor environment are more challenging as compared to outdoor environments. Different sensors and signals have been utilized by researchers for position tracking and indoor navigation. There are a few important aspects of a typical indoor navigation system that are usually considered by the community and researchers. These aspects include implementation cost, ease of use, ease of installation, efficiency, accuracy, and portability. The camera is a low-cost solution that has been used as an alternative to the human eye (for blind users) or as additional support (for normal or visually impaired people). Unlike sensors-based or other alternatives such as networking methods, computer vision provides a low-cost, simple, and easy-to-use solution for indoor navigation. Computer vision-based methods are generally utilizing a video camera (fixed or mobile) to assists users to navigate in unfamiliar environments. Vision-based indoor navigation is either implemented by placing additional landmarks in the scene or by using indoor images via their natural features. The landmarks (e.g. fiducial markers) are more robust than natural features, however, adding these landmarks are often difficult to install and the building administrators (owners) do not like to place additional images. The key issues with the vision-based method are robustness in different light intensities and occlusion of the landmarks.

Tables 1 and 2 present a summary of our extracted data from the selected articles. The key input to a vision-based system is the camera, whereas the feedback is audio or visual or a combination of both. Fig. 9 shows a histogram plotted on the basis of publication year, where we can see a rapid increase in recent years. Vision-based methods have been categorized based on the way they are used and the tools they use. Each category has its own advantages and limitations. Marker-based methods are superior in robustness over markerless methods. The portability and ease of use are also important. In this regard, smartphone based methods are usually preferred over those methods that used laptops or other devices. Similarly, wearable devices are also found with a significant improvement in user satisfaction. Wearable devices provide haptic feedback and are therefore more realistic and helpful for blind users. Moreover, these devices are easy to carry during navigation than hand-held devices (such as a smartphone). However, wearable technology is difficult to install and requires additional costs. Hybrid systems are superior to all other methods and often provide highly robust solutions. This is because the hybrid systems use additional sensors with computer cameras to provide an alternative solution whenever the camera may somehow fail to detect a target object or image.

The way the feedback is given to the user is also important for usability analysis. Computer vision-based methods are mostly found with visual and/or audio feedback. The visual feedbacks are typical with realistic AR overlays over the indoor scenes. AR provides a more realistic solution that enhances users' satisfaction and trust. Audio feedback is usually preferred for blind and visually impaired people. Audio instructions are helpful for turn-by-turn navigation. Systems with more than one type of feedback (e.g. audio and visual) are more user-friendly. Tactile feedback is also helpful for blind and VIP.

Results analysis and system evaluation also play a significant role in research. There are four general types of evaluations in indoor navigation:

1. Some systems are evaluated subjectively via a user study with the main focus on usability analysis and user satisfaction.
2. Some systems used different simulations for evaluation.
3. Other systems used some existing dataset for the result analysis.
4. Few articles have used more than one method of evaluation.

According to our study, 31 out of 68 selected articles evaluated their systems with a user study i.e. 45.6% of the total. There are different metrics used for the evaluation of indoor navigation methods. Few of these metrics include accuracy, efficiency, ease of use and implementation, and implementation cost.

The selected articles were analyzed for quality assessment. Fig. 10 shows a chart of the articles' quality assessment. If we compare the quality score with a similar study in other field [27], the vision-based indoor navigation did not reach high-quality articles as the scope of the study is very narrow. Even though the primary search gave a relatively higher number (*i.e. 5050* documents) as compare to *4074* documents in the above-mentioned SLR [27]. Multi-sensor fusion solutions are found comparatively more robust and with higher accuracy. For example, *ARBIN* [89] which uses Google ARCore [143] and *BLE* beacons which achieves an accuracy of 97%. Similarly, another fusion of camera and depth sensors [93] achieved 99% accuracy. Similarly, the fusion of marker or other landmark detection with depth sensor is another robust and accurate system [125]. However, the cost of additional sensors may increase the cost. A number of methods and datasets can be found in a recent study [191] on the fusion of multi-sensor approach.

7.1. Limitations of this study

Though we have used SLR for article searching and inclusion/exclusion criteria. Still, due to too many articles on the topic, it is possible, we might have excluded some relevant articles. The categorization of articles is also overlapping each other. Some articles may belong to more than one category (for example, smartphone-based method and marker-based method). Since we have shortly described each individual article, there might be a little lack of continuity or few repetitions (for example in describing limitations of different articles).

7.2. Conclusion

Indoor navigation has remained an active research topic for the last 10 years. Several primary studies, as well as secondary

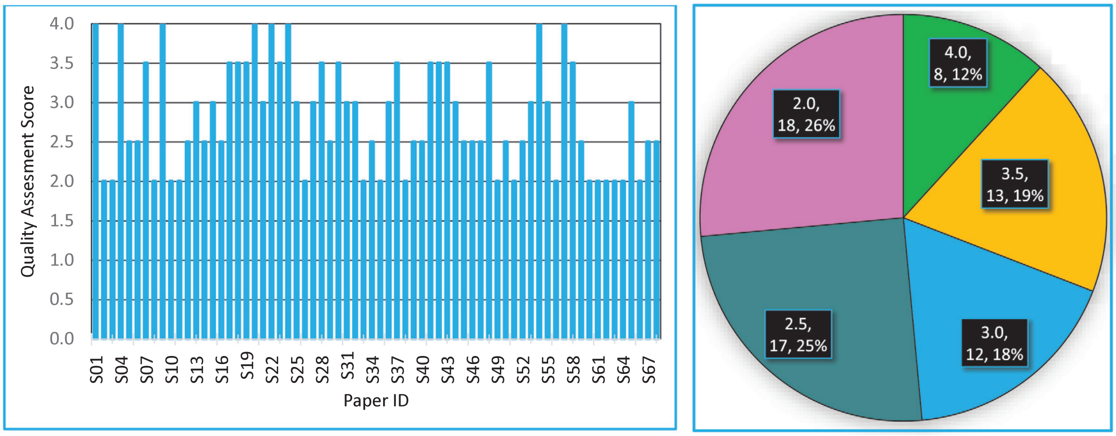

Fig. 10. Article Quality assessment: Left: Distribution of the articles on the basis of their quality assessment. Right: Pie chart showing percentage values of article in each category of the quality assessment (i.e. Quality assessment score, number of articles, and percentage value).

studies (reviews of primary studies), have been published time by time. Vision-based indoor navigation is a specific domain providing interesting low-cost solutions to the community. In this paper, we followed a systematic literature review methodology and reviewed state-of-the-art vision-based indoor navigation methods. In total, we selected 68 articles via SLR methodology. We analyzed these articles critically with their advantages and limitations. We categorized the articles into different categories and also suggested a few interesting future research gaps. Each article was examined for its quality assessment. In the future, we are planning to design an algorithm that automatically creates a 3D model of a given building that can be used for on-site indoor navigation as well as a virtual tour from a remote area

CRediT authorship contribution statement

Dawar Khan: Initiated the project, Wrote the first draft, and then revised it on mutual disucssions with other authors. . **Zhanglin Cheng:** Supervised the project, Helped in article revision, Guided the organization, Methodology, Gave comments on various aspects of the review processing, Article writing. **Hideaki Uchiyama:** Helped in article revision, Data collection, Review methodology. **Sikandar Ali:** Helped in article revision, Data collection, Review methodology. **Muhammad Asshad:** Helped in quality analysis article search, Revising the article. **Kiyoshi Kiyokawa:** Supervised the project, Gave comments on various aspects of the review processing, Article writing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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